YUVA YODHA ENERGY TECH HACKATHON 2026 · CHALLENGE 04: SMART MANUFACTURING

MeltWise

Every melt, measured.

An AI energy copilot that helps India's induction-furnace foundries use less power per tonne, with the same castings.

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THE PROBLEM

Foundries use more power than they need

~5,000
foundries in India; 90% are MSMEs

15.86 Mt
castings made in 2024–25; world's #2

13–21%
more melting energy than industry norms

15–30%
of production cost can be energy

And the cost of that waste is rising

- Time-of-day tariffs since April 2024: solar hours 10–20% cheaper, peaks 1.2× or more
- Global buyers now ask suppliers for carbon data on what they buy
- Most SMEs sit outside PAT norms and have no energy manager

Sources: The Machine Maker 2025 · foundry energy study (ResearchGate) · challenge brief · MoP ToD Rules 2023

ROOT CAUSE

The waste hides inside every melt

Today's melt: 75 minutes



~650 kWh/t

With MeltWise (likely case): 62 minutes



~617 kWh/t

Melting at full power · Part-power melting · Holding while moulds wait · Time and energy saved

50–80 kWh/t

swing between melts in one furnace (BEE cluster study). Monthly bills can't see it.

Loss rates in our model

- Overshoot: ~3 kWh/t per 10 °C above spec
- Holding or slow melting: ~1 kWh/min (1 t furnace)
- Lid left open: ~1 kWh/min (1 t furnace)
- Sand or rust in charge: ~1.1 kWh per kg

Sources: BEE SME Programme foundry-cluster DPR · Foundry Management & Technology · MeltWise impact model

THE SOLUTION

MeltWise: one loop, run on every melt

01

Measure

Reads the furnace panel meter every second. Operators log charge, scrap mix and temperature.

02

Detect

A physics + ML model predicts the kWh this charge should need and names the cause of any excess.

03

Guide

Live prompts at the furnace, in the operator's language: close the lid, add power, tap now.

04

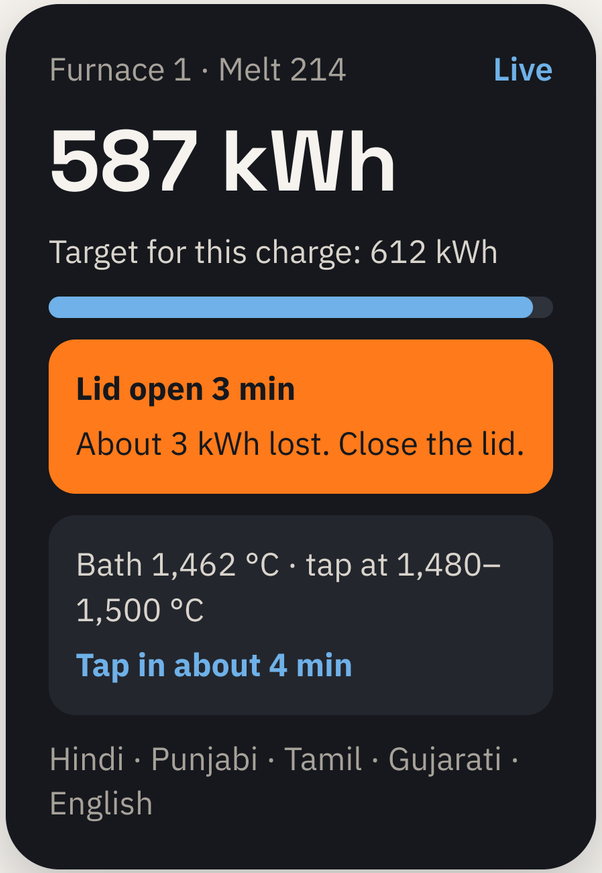
Plan & report

Schedules melts around the moulding line and solar-hour tariffs. Monthly savings and CO_2 reports.

Offline-first edge gateway · ₹87k kit per furnace · nothing on the furnace changes · reads Schneider EM6400NG and PowerLogic meters

USER JOURNEY

Built for the people at the furnace



Furnace operator

Live prompts in their own language: close the lid, add power, tap now.

Shift supervisor

A melt plan synced with the moulding line, and alerts when metal sits holding.

Foundry owner

Monthly kWh/t trend, rupees saved and tariff savings, on WhatsApp or PDF.

Buyer or OEM

A per-batch record of kWh/t and CO_2/t, aligned with the GHG Protocol.

Left: operator app mock-up. The same prompts also appear on a shop-floor screen.

ARCHITECTURE & TECHNICAL APPROACH

Edge-first, built for the foundry floor

ON THE FURNACE

Furnace sensors

- Panel meter, Modbus, 1 reading/s
- Lid switch, thermocouple (optional)
- Operator app: charge, scrap, grade

IN THE FOUNDRY

Edge gateway

- Works offline, stores 7 days
- Splits each melt into phases
- Runs the melt model locally
- Drives shop-floor prompts

SYNCS WHEN ONLINE

MeltWise cloud

- Physics + ML: expected kWh per charge
- Cause finder for excess kWh
- Scheduler: moulds + ToD tariff
- M&V and carbon (CEA factor)

FOR PEOPLE

Apps & systems

- Operator app, shop-floor screen
- Manager dashboard (web)
- Owner, buyer reports (WhatsApp, PDF)
- Excel, Tally, ERP sync; open API

Stack: Python (pandas, LightGBM, OR-Tools) · FastAPI · TimescaleDB · MQTT · React PWA · Modbus

Data model: Furnace → Melt → Phase → Reading, plus Charge, Alert, Schedule and Batch report

Schneider fit: reads EM6400NG / PowerLogic meters; feeds EcoStruxure or AVEVA over Modbus / OPC UA

INNOVATION

What MeltWise does that others don’t

What it does	Monthly bill	kWh integrator (BEE)	Generic IoT dashboard	MeltWise
Energy for each melt, not each month	No	Yes	Partly	Yes
Says why a melt used extra energy	No	No	No	Yes
Live prompts in the operator's language	No	Number only	No	Yes
Plans melts around moulds and tariffs	No	No	No	Yes
Per-batch CO_2 record for buyers	No	No	Partly	Yes
Works offline, under ₹1 lakh per furnace	n/a	Yes	Rarely	Yes

Built for small foundries: local-language prompts, an offline edge, no changes to the furnace, and payback in months.

Fits Schneider Electric's stack: reads EM6400NG and PowerLogic meters; a natural next step is Altivar drives on cooling-water pumps.

Comparison based on the BEE cluster DPR (kWh integrator) and typical IoT monitoring products

IMPACT VS A MEASURED BASELINE

What one foundry gains in a year

33.2
kWh less per tonne melted (5.1%), likely case

₹9.45
lakh net saving a year, after subscription

56.6
tonnes of CO_2 avoided a year

1.1
months to pay back the kit (4.9 if conservative)

Per foundry, per year	Conservative	Likely	Stretch
Energy saved per tonne	16.7 kWh (2.6%)	33.2 kWh (5.1%)	42.9 kWh (6.6%)
Net saving after subscription	₹3.70 lakh	₹9.45 lakh	₹14.75 lakh
CO_2 avoided	28.5 t	56.6 t	73.1 t
Payback on the kit	4.9 months	1.1 months	0.7 months

At 1,000 foundries (likely case): **80 GWh, ₹64 crore and 57,000 t CO_2 a year**

Baseline: 1 t furnace, 2,400 t/yr, 650 kWh/t, ₹8/kWh, CEA 0.710 kg CO_2/kWh; tariff savings are cost only

FEASIBILITY & BUSINESS MODEL

Affordable for a small foundry

What a foundry pays

Item	Cost
Modbus energy meter (EM6400NG)	₹17,289
CTs, wiring and enclosure	₹15,000
Edge gateway	₹20,000
Shop-floor screen or tablet	₹15,000
Installation and training	₹20,000
One-time kit per furnace	₹87,289
Subscription per furnace	₹4,000/month

Pays back in 1–5 months

Who we sell to

Induction-furnace iron foundries, 500 kg to 5 t. ~5,000 units in India, 90% MSMEs. Start with two clusters.

How we reach them

Through foundry cluster associations: a demo furnace, a local installer, results shared with members.

How it’s paid for

Owners pay from savings, or shared savings for cash-tight units. MeltWise data backs SIDBI 4E upgrade loans.

Risks and answers

Adoption: local-language prompts. Internet: offline edge. Trust: savings checked on the plant's own melts.

Prices: Moglix/IndiaMART listings (meter), indicative quotes (rest) · SIDBI 4E: loans up to 90% of project cost

ROADMAP

From prototype to 1,000 foundries

OCT–NOV 2026

Prototype

Melt simulator (physics + ML) tuned on published cluster data, plus operator app, dashboard and scheduler.

Target: predict melt energy within 5%

JAN–JUN 2027

Pilot

3–5 foundries in one cluster: 4 weeks of baseline melts, then 8 weeks with MeltWise, checked independently.

Target: 17+ kWh/t saved, verified

2027–28

Cluster rollout

50 foundries through an industry association, with local installers and a SIDBI 4E tie-up for upgrades.

Target: payback under 6 months

2028 ONWARD

Scale

1,000 foundries, then steel induction melting, aluminium and forging furnaces, plus predictive maintenance.

Target: 80 GWh and 57,000 t CO_2 a year

Our ask: mentors in industrial energy, meters for the pilot, and introductions to foundry clusters.

TEAM

The team behind MeltWise

SS

Surianandhan Sridhar

Team lead

SH

Sameera Hussainy Pattan

Team member

Together we cover Python, machine learning and data science, and web and app development.

MeltWise · Every melt, measured.

Yuva Yodha Energy Tech Hackathon 2026